

CHAPTER 2

LITERATURE SURVEY

Image brightness is non uniform during the time of image acquisition. Certain parameters and factors tend to produce variations of the brightness in the absence of image values which are random without any standard pattern. The random variation in image brightness is designated as noise. Noise elimination plays an important role in image processing. Images are generally added by noise when they are acquired, transmitted and processed. The noisy images need to be manipulated in such a way that it becomes a noise free component, either for gray scale or colour images.

2.1 NOISE MODELING

Mandar *et al.* (2015) explains that development in digital technologies, the image capture devices such as digital cameras and computers becomes more affordable price. Since digital images were generally used in research fields, a preferable care sustains the quality. Unfortunately, there are possibilities of corrupting the image unintentionally which are called as Noise. The noise occurs at the time of acquisition, due to transmission, atmospheric condition, electrical fluctuations or malfunctioning of image capture device. The various types of noise model which adds in the images are classified as Spatially dependent noise such as Periodic noise and Spatially independent noise models such as Gaussian Noise, Rayleigh Noise, Gamma Noise, Exponential Noise, Impulse Noise.



Removal of noise from the digital acquired image is unavoidable due to huge intervention of electrical machineries, atmospheric condition, status of the objects etc., A detailed review on noise modeling was studied and a quantitative analysis of the noise models were also discussed by Ajay Kumar Boyat *et al.* (2015).

Etal Gravel (2004) studied noise in various medical images through statistical properties with an appropriate method based on noise types. Generally, it was assumed that the noise is an additive source with zero mean, constant variance, Gaussian distribution for various types such as photon, electronics and quantization. It was also suggested that the noise variance for non-linear function could be modeled better when the image intensity value depends on external parameters. A method was proposed and applied for MRI and various types of X-ray images that relates the image intensity and noise variance.

The conventional methods of filtering, Linear and non-linear, removes the noise based on the nature of image corruption. Different approaches will produce comparative results. Linear filters are not suitable for dealing impulse noise, rather non-linear filters, which preserves edges and smoothen the images. The quality of image improves when median filters are used for noise removal without destroying any variation in the colour component studies by Abdalla Mohamed Hambal *et al.* (2017).

2.2 IMAGE NOISE DETECTION

Maitham D Naeemi *et al.* (2016) described that in pathology detection and procedural treatments, CT images are widely used for accurate identification of diagnostic. A novel approach based on Fourier Transform metric for image quality estimation was proposed. By considering the phantom and patient CT images using variance, standard deviation etc., is



directly proportional to the tube current. But combining phantom and patient CT, the variance and standard deviation do not correlate.

The outlier reduction was performed by Christy *et al.* (2015) to sustain the concept of data pre-processing in which two algorithms called Distance based outlier and Cluster based outlier algorithms were introduced as it was experimented 10% data may be erroneous or missing or not available. . These algorithms were utilized for detecting and removing outliers using an outlier score. By performing certain tests it was proved that the cluster based outlier detection gives better accuracy than the distance based outlier detection method.

Ahmed M Elmogy *et al.* (2017), studied image denoising as the major task when high level noise suppresses the quality of the image. GridLOF algorithm removes the harshness of the image by following a method called as Enhanced GridLOF algorithm. This technique was suggested for minimizing the time and complexity of algorithm and it was proved by its simulation results.

Considering the outliers, Breunig *et al.* (2000) data not relevant to useful information which is an equivalent word preferred to indicate noise focuses relative data density with a probability and not against the neighbours.

For enhancing the LOF algorithm and to perform better in outlier detection, Mahmoud *et al.* (2016) performed Machine Learning and Artificial Intelligence algorithm and many other research activities were also performed.

Das & Rasmi (2015) prefer detecting noise from the affected portion of the image with an objective of recovering or reconstructing the noiseless image. Following this, noise was initially detected which was then



made ready for applying for noise removal by suitable algorithm which concentrates on the affected pixels.

Zhou *et al.* (2012) proposed the modern methods such as non-local mean, PDE based method for noise removal. But the proposed methods modify the noisy and non-noisy pixels

The modification of noisy and non – noisy pixels of the image extended the researchers Elad & Aharon (2006) performed the noise removal through fuzzy logic which depends on fuzzy impulse detection for noise detection without advancement in training.

Neuro-fuzzy operator was also proposed by Kindermann *et al.* (2005) which is based on two adaptive Normal Filters, which fix the noisy pixel and rectifies the right pixels. This filtration process was carried over with a post –processor.

The LOF follows a hierarchical framework for outlier detection with effective performance. But the complexity of identifying the nearest neighbours in LOF algorithm was difficult which made much of researchers like Immerkaer (1996) to skip the deducing stands of nearest data point in the image.

2.3 NOISE ESTIMATION

The noise measurements and removal in images are performed by Zhifei Zhang (2016) through analysing the statistical details of various approaches, such as filter based, block based and wavelet based for measurement. The spatial domain, transform domain, TV based minimization, and Sparse coding techniques are used for noise removal. Progressive result



was achieved through Grouped Sparse Coding. It has been focused to retain the details of images even the noise removed.

Walid Sharabati & Bowei (2018) studied the noise reduction in image after using various algorithms for retrieving the input image as noise free for interpretation and analysis. The author attempted to get the performance of various filtering techniques such as Median, Average and Wiener after removing the noises like Gaussian, Poisson, Speckle and Salt and Pepper. The results are evaluated and compared for entropy and processing time.

In image processing, segmentation and object recognition requires de-noised image. Filtering techniques are focussed by Qin Xu *et al.* (2018) on removing single or multiple noises. Considering the neighbourhood regression approach, multiple types of noise such as Gaussian, Poisson, Mixed Gaussian and Poisson, Salt and Pepper and Stripped noise are predicted. Parallelized technique, filters which do not requires any prior knowledge of the variance to remove the noise from image. The filtered output image is compared with other popular de-noising methods such as BM3D, Median filters and wavelet filters showing that regression filter performance is better than other conventional methods.

Chithra & Santhanam (2018) acquired the image in real time from different devices such that CCTV, CT scanner, MRI scanner, X-rays, Ultrasound etc., are naturally added by multiple noises. Absolute difference, mean and median methods were applied to remove the Salt and pepper and Gaussian Noise combination and Speckle noise combination. From the performance analysis, the above method provides better result comparing to Fuzzy based filter (FBF) and Median Weiner Bilateral Filter (MWBF). But the method is restricted to JPEG format for gray scale and colour images.



Priyan Chatterjee & Peyman Milanfar (2010) studied the image acquisition and concluded that the nature of noise study is important to elaborate and implement the de-noising methods. It is identified that the image de-noising is a well studied problem. But the researchers are performing different techniques to remove the noises from the images proving that the de-noising is not completed.

Kui Liu *et al.* (2014) proved that Spilt Bregman method is an improved technique comparing to other methods to avoid staircase artifacts and eliminates noise from the image by weighted combination of Tikhonov regularization and Total variation regularization to maintain smoothness and preserve edges and corners with highest PSNR and structural similarity index measure (SSIM). But the image size is restricted to 256x256.

Image de-noising is an universal problem of noise removal is performed by different methods. Patch based image de-noising method gives the most optimistic performance comparing to other methods to the state of art results. By comparing the edge and smooth regions, finite ridgelet transform are applied to preserve better directional singularities. The output quality of this ridgelet transformation has the least artifacts in the images and suits for natural images.

Deledalle *et al.* (2009) proposed an enhanced version of non-local means algorithm which was used in calculating weighted average values of similar patches. The noise removal process has been expressed as weighted maximum likelihood estimation problem, where the weights were derived in a data driven way iteratively and refined on noisy patches, similarity of patches extracted from previous estimate for Synthetic aperture radar (SAR) images.

Luo & Enming (2016) studied the ill posed problem in image, a detailed statistical analysis and calculation has been made to utilize the



default images for exploring their effectiveness with prior knowledge of noises. By introducing adaptive patch based de-noising algorithms such as, adaptive non-local means for multi-view image de-noising, adaptive image de-noising and adaptive image de-noising by mixture adaption noise removal is performed but resulting images derived for inverse problems such as deblurring, repainting and super resolution.

Interpretation of the images for de-noising was performed by Dai-Viet Tran *et al.* (2017) by patch based approach, image formation model and prior image distribution was obtained by incorporating Bayesian framework. By extracting the probability directly for the distributed image patches, authors has suggested that the image prior is more efficient than the sparsity model of noise removal which is expected to be the state of the art.

Kervrann & Boulanger (2006) had proposed the representation of image de-noising as a point wise selection of small image patches which is of constant size and dynamic neighbourhood. Contributing the sum of data point within the adaptive neighbourhood, the accuracy of approximation and the stochastic error will be balanced at spatial position. This method is proposed to work with parameter free algorithm and applicable for artificially corrupted and real images showing that the performance is better than other existing methods but not evidencing to the level of ground truth.

In computer vision, sparse learning techniques became popular where the improvement has been undergone by Charles-Alban Deledalle *et al.* (2011) in visualization. The patch based de-noising methods proposed, nearing the state of the art can be achieved using orthogonal dictionaries which are classified as local Principle Component Analysis (PCA), Hierarchical Principle Component Analysis and Global Principle Component Analysis.



It has been stated by Monagi *et al.* (2017) proposed patch based image de-noising methods like BM3D, Non Local Mean, Principle Component Analysis (PCAP method and K-Singular Valued Decomposition (K-SVD) method produce comparative results among other conventional methods of de-noising. The proposed NL mean technique performs only for low noise level. K-SVD is expensive and fails to compete with BM3D technique even though the sparsity is learned from the data. It has been concluded that BM3D works better for all noise levels and all images with lower execution time.

Priyam Chatterjee & Peyman Milanfar (2012) had developed and implemented a Patch based Locally Optimal Wiener (PLOW) filter based de-noising algorithm with clear statistical motivation and analytical formulation without any parameter tuning. The method is proved to be effective for grayscale and colour image de-noising as well and the results are compared with the existing algorithm. It has been proposed that the de-noising method can be improved by correlating the colour components. It is also proved that, PLOW shows comparable results with BM3D and Nonlinear Linear Square Method (NLSM). Though in this method geometrically similar patches were proposed, clustering steps may be considered on colour images and the noise were assumed to be homogenous and suitable filter can be applied for noise removal.

In the patch based Wiener filtering technique, by exploiting the patch redundancy both geometrically and photo metrically, similar patches has been used to estimate different filter parameters by Dharshini & Sivaranjani (2013). The input noisy image has been estimated directly and proved that the de-noising method can be applied in colour image for noise removal which may corrupt the RGB colour space conversation. K-mean clustering technique has been initially proposed for noise removal and



proposed the locally optimal Weiner filter method for patch redundancy and showing the filter performance.

Gadde Yasasvi & Durgaprakash (2017) considered the image as a global pattern and identified that there is possibility of image blurring or damage. By providing aqua ring noise portion, the segmented or grouped pixel performs de-noising by clustering the image and applying Gaussian Mixture Model (GMMS) to improve the image quality under severe noise environment.

For patch based de-noising, clusters of noisy images of unique geometric structure are considered by Priyam Chatterjee & Peymann Milanfar (2009). It has been proposed K-LLD method, a locally adaptive de-noising technique which is enhanced to a novel mechanism for optimally choosing the local patch size for each cluster using Stein's Unbiased Risk Estimator (SURE) which is highly competitive with other recent state of art methods. For obtaining optimized results, K-means, mean shift methods are also recommended.

Jerome Gillies *et al.* (2014) proved adaptive wavelets will be practically flexible by considering different options to execute the Fourier boundary detection. The expansion of the functions varies based on the detection task and noise has been considered as the most typical step in the EWT and the detection depends on the specific analysed image. Though these transforms were preferred and it provides sparse representation and helps in restoration, the algorithm implemented will be selected if the Fourier Spectrum has or not clear distinct modes. By implementing some noise removal transforms and algorithms, it proves to extend the method for advanced noise removal methods.



Ungsumalee Suttapakti *et al.* (2017) described in detail on monocomponent image decomposition. The former monocomponent methods performance was inaccurate because of interference and redundancy component problems, because of inaccuracy in spectrum segmentation. The proposed Empirical Monocomponent Image Decomposition (EMID) performs for better results by energy concentration based segmentation and developing of empirical image filter bank. This procedure produced an accurate spectrum segmentation and reduces the interference and redundancy component. The image decomposition is performed based on energy concentration in Fourier support. This EMID technique was suggested for noise detection, noise removal in a better way.

2.4 NOISE REMOVAL

Thakur Kirti *et al.* (2015) focused on removing Poisson Noise from X-ray images instead of increasing the dose value which might affect the patient. The author used a modified Harris Corner point detector to predict noisy pixels and proposed a median filter and moving average filter for the noise removal. Experiment result of the proposed method is ideally equivalent to the state of art method called as non-local means Poisson noise filter. The method introduced was simple with good visual quality and less execution time. Translation invariant two dimensional Dyadic Wavelet Transform (DyWT) was designed by Pierre Vandergheynst (2002) in a simple and efficient method which relies on dyadic decomposition, which is constructed from Continuous wavelet transform. Since, the method has the advantage of designing 2D with lot of relaxations, construction of directional wavelets such as Quadrature Mirror Filters (QMF) are possible and whose filtering capabilities are having much importance in image processing techniques. A new technique was introduced to design and implement 2D translation which allows an effective representation of the objects. The technique was very successful in texture analysis and synthesis, watermarking and image quality assessment.



In medical imaging, precise diagnosis was suggested by Mohd Ameen & Shah Aqueel Ahmed (2016). To practice proper treatment, image denoising is essential. The author proposes that for precise analysis, the image capturing methods like X-ray, CT, MRI, PET, SPECT etc., would lose pixel in certain regions. This might create misdiagnosis during treatment. Meticulously to avoid this sort of threat, denoising process is to be performed. Linear Minimum Mean Square Error Estimation technique was proposed as an ideal method for denoising.

Mirajkar Pradnya (2013), outlined Image fusion as the procedure of mixing 2 or more of in contrast to images into a brand-new single image holding their main features from every part of the pictures with a detail's information. Spatial fusion and Rework fusion are introduced and utilized to perform the image fusion. The associated degree image fusion was supported by stationary ripple transform, which was applied foremost to initial image in order to induce the sting image. The area of edge pictures is in unit and combined to get the entire edge image exploitation spatial frequency measurement, which is comparable with easy fusion strategies.

Hindu deity Kumar *et al.* (2013), studied a picture resolution improvement technique which supports interpolation of high frequency sub band pictures obtained by separate ripple rework (Discrete Wavelet Transform - DWT) and also the input image. The nursing input is applied to decompose and change it to different sub bands. The resulting high frequency sub bands exploits through Stationary Wavelet Transform (SWT). The square measure combined to generate a brand-new high-resolution image which exploits Inverse Discrete Wavelet Transform (IDWT). The quantitative results are with visual square measure and show the prevalence of the planned technique over the conventional image enhancement method.



Siva Kumar *et al.* (2013), proposed a method based on the interpolation of the high frequency sub band images obtained by DWT, the enhancement in resolution of image is performed. The images are decomposed by DWT and the edges are extracted by Stationary Wavelet Transform (SWT) resulting in different sub bands of the input. The high frequency sub bands are modified by interpolating the sub bands with input images. The quantitative and visual results show fine output compared to that of the conventional state of art enhancement techniques.

Kanagaraj Kannan *et al.* (2010) introduced the fast development of digital image process algorithm which results in the growth of feature in image fusion. The method of combining 2 completely different pictures into a single image by retentive salient options from every image, with extended pixel content is through image fusion. Two approaches of image fusion are spatial fusion and transform fusion. Discrete Wavelet model plays a vital role in image fusion since it minimizes structural distortions among the assorted alternative transforms. Lack of shift in modifications, poor directional property and also absence of section data are the drawbacks of the Discrete Wavelet model. These drawbacks are overcome by Stationary Wavelet Remodel and twin tree advanced wavelet remodel. The work describes the optimum decomposition level of distinct, stationary and twin tree advanced wavelet remodel required for better fusion constituent in terms of Root Mean Square, PSNR, Quality Index.

Amar Tej (2015), proposed a preprocessing technique for enhancing the medical image resolution with good quality. To preserve the edges and contour information, efficient denoising method was suggested such as Average filtering, Median filtering, Wiener filtering and also by wavelet denoising. The author also suggests Standing Wave Transform and DWT for noise removal and parameters such as PSNR, is evaluated for the performances of algorithm.



Ashish goud Purushotham (2015), shows fusion results in a new image, which suits for human and machine perception. Pixel level image fusion using wavelets and principal component analysis have been utilized and the performance metrics were observed with and without reference images that concluded image fusion using decomposition of higher order with better performance by PCA compared to other methods. But it was identified that DWT performs better than PCA.

2.5 INFERENCES FROM LITERATURE SURVEY

From literature survey, it is inferred from various work on image that the noise removal is continuously explored for more clear vision specifically in CT images. Research articles leads to the identification of various causes of noise added in CT image.

1. From the literature survey the noise detection methods available in image processing are supposed to be time consuming, complex and tedious which made the researchers to skip the step mostly. The noise estimating procedures surveyed are also tedious and the processing time for estimation noise is found to be high.
2. The acquisition of such noisy images, are followed by detection of noise and then estimating the noise level in the image both globally and patch-based techniques are required.
3. The estimation is performed by calculating various statistical parameters such as mean, variance, standard deviation and process time for its optimization. The noise estimated image are filtered by an appropriate method and makes suitable for proper diagnosis as noise free images are mostly required.

